

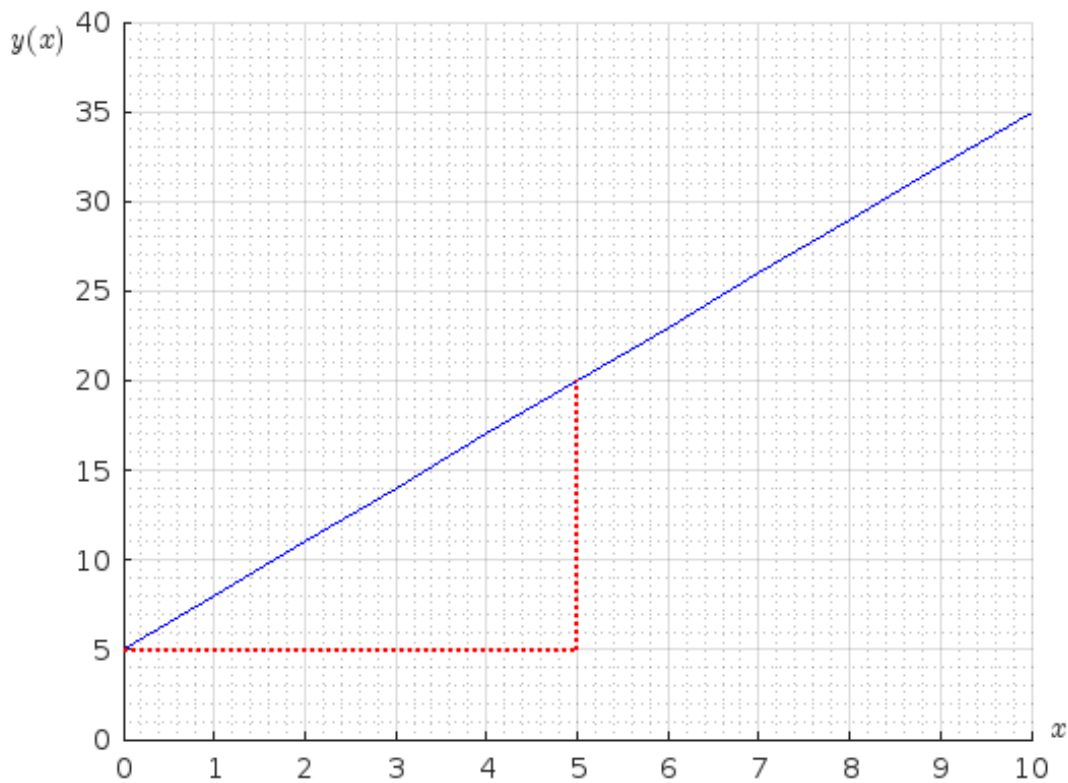
Rail Engineering Mathematics

Tutorial Sheet 1

Solutions

1. Determine the equation of the line from the following graphs:

(a)



Solution:

In each case, we state the general equation of a straight line $y = mx + c$, then determine the y -intercept (c) from looking at the height of the graph where it crosses the y -axis (at $x = 0$). The gradient (m) is then determined by Δy over Δx .

For part (a):

The equation of a straight line is $y = mx + c$.

Then since the graph crosses the y -axis at $(0, 5)$, we can determine that $c = 5$.

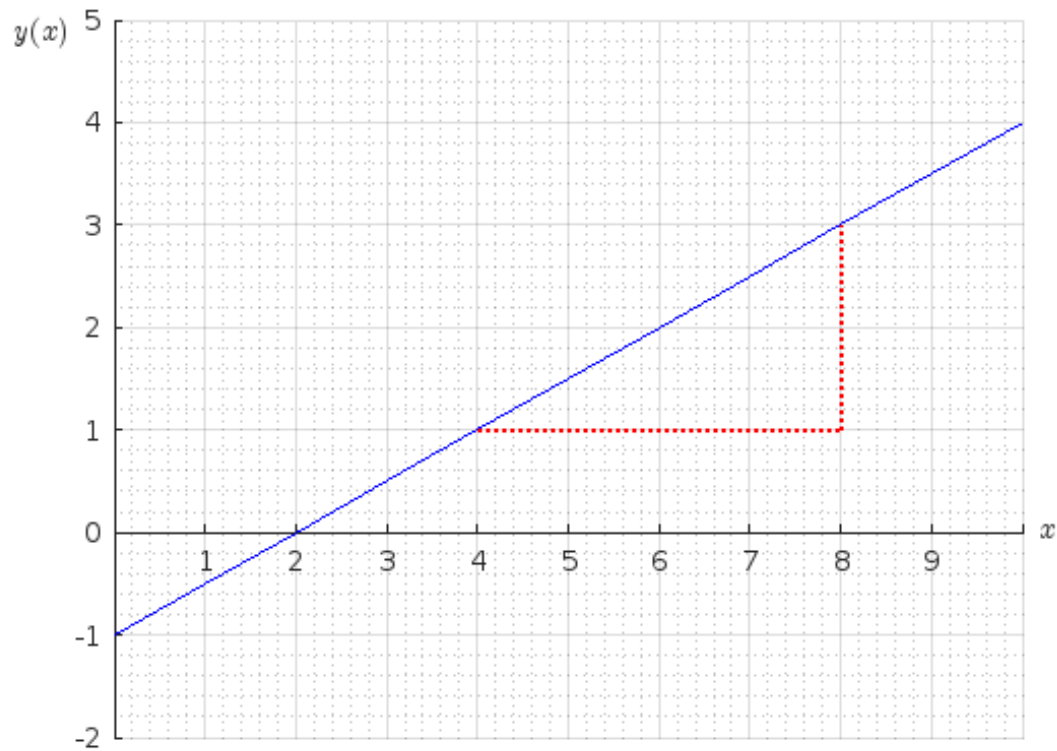
Choosing this point, with $x_1 = 0$ and $y_1 = 5$, and a second point $(5, 20)$ that the graph passes through so that $x_2 = 5$ and $y_2 = 20$, then the gradient m is given by:

$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} \\ &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{20 - 5}{5 - 0} \\ &= \frac{15}{5} \\ &= 3 \end{aligned}$$

Hence:

$$y = 3x + 5$$

(b)



Solution:

The equation of a straight line is $y = mx + c$.

Then since the graph crosses the y -axis at $(0, -1)$, we can determine that $c = -1$.

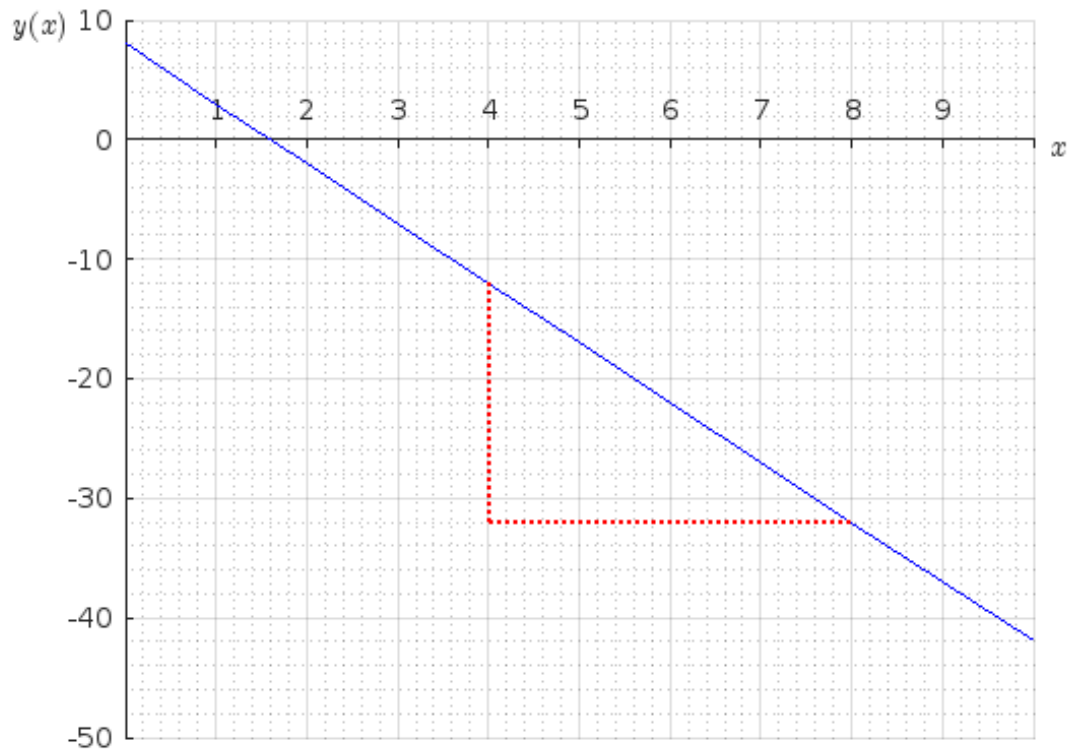
Choosing two points $(4, 1)$ and $(8, 3)$ that the graph passes through so that $x_1 = 4$ and $y_1 = 1$, and $x_2 = 8$ and $y_2 = 3$, then the gradient m is given by:

$$\begin{aligned}
 m &= \frac{\Delta y}{\Delta x} \\
 &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{3 - 1}{8 - 4} \\
 &= \frac{2}{4} \\
 &= \frac{1}{2}
 \end{aligned}$$

Hence the graph has equation:

$$y = \frac{1}{2}x - 1$$

(c)



Solution:

The equation of a straight line is $y = mx + c$.

Then since the graph crosses the y -axis at $(0, 8)$, the y -intercept is $c = 8$.

Choosing two points $(4, -12)$ and $(8, -32)$ that the graph passes through so that $x_1 = 4$ and $y_1 = -12$, and $x_2 = 8$ and $y_2 = -32$, then the gradient is:

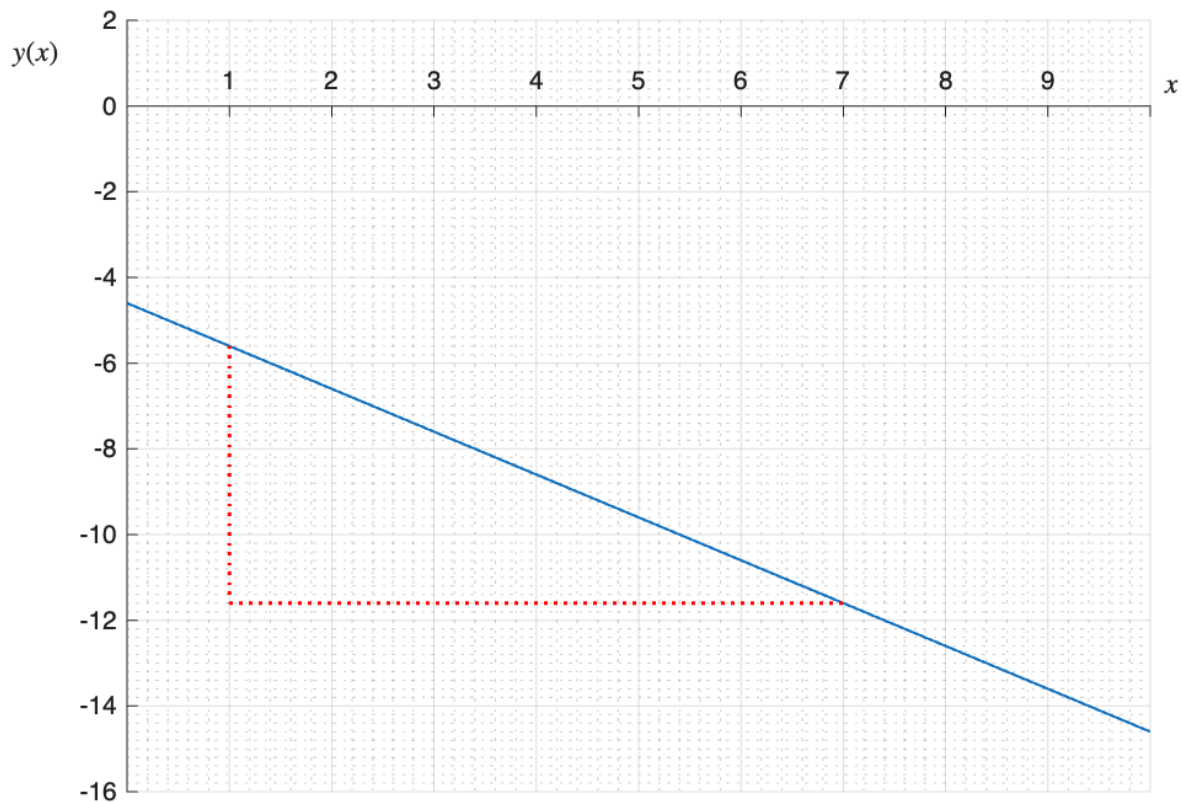
$$\begin{aligned}
 m &= \frac{\Delta y}{\Delta x} \\
 &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{-32 - (-12)}{8 - 4} \\
 &= \frac{-20}{4} \\
 &= -5
 \end{aligned}$$

Of course, a negative gradient is expected as the graph is sloping downwards.

Hence:

$$y = -5x + 8$$

(d)



Solution:

The equation of a straight line is $y = mx + c$.

Then since the graph crosses the y -axis at $(0, -4.6)$ (note that each box on the y -axis denotes an increment of 0.2), the y -intercept is $c = -4.6$.

Choosing two points $(1, -5.6)$ and $(7, -11.6)$ that the graph passes through so that $x_1 = 1$ and $y_1 = -5.6$, and $x_2 = 7$ and $y_2 = -11.6$, then the gradient m is:

$$\begin{aligned}
 m &= \frac{\Delta y}{\Delta x} \\
 &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{-11.6 - (-5.6)}{7 - 1} \\
 &= \frac{-6}{6} \\
 &= -1
 \end{aligned}$$

Hence the equation of this graph is:

$$y = -x - 4.6$$

2. Sketch the graph of the following functions, indicating the points at which the line crosses the x and y axes:

(a) $y = 3x + 2$

(b) $y = 3x - 2$

(c) $y = 3x$

(d) $y = 1 - 2x$

(e) $y = -x + 1$

(f) $y = 1$

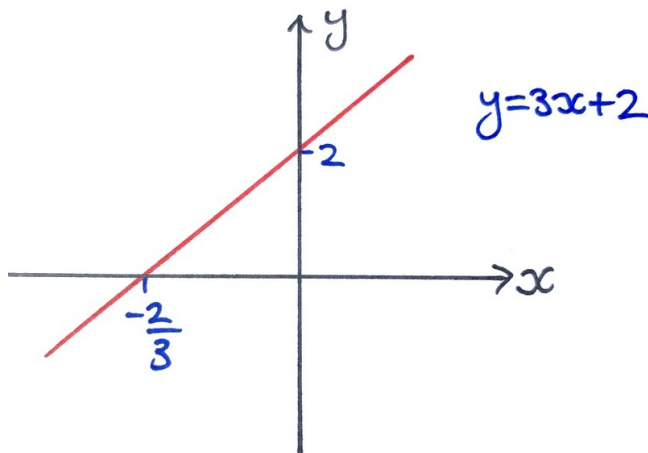
(g) $y = 1 + 2x$

Solution:

In each case, the easiest way to do this for a straight line is to determine the y -intercept, and the x -intercept (i.e. the value of x when we set $y = 0$). Any two points are sufficient to define a straight line, so then we just join them up.

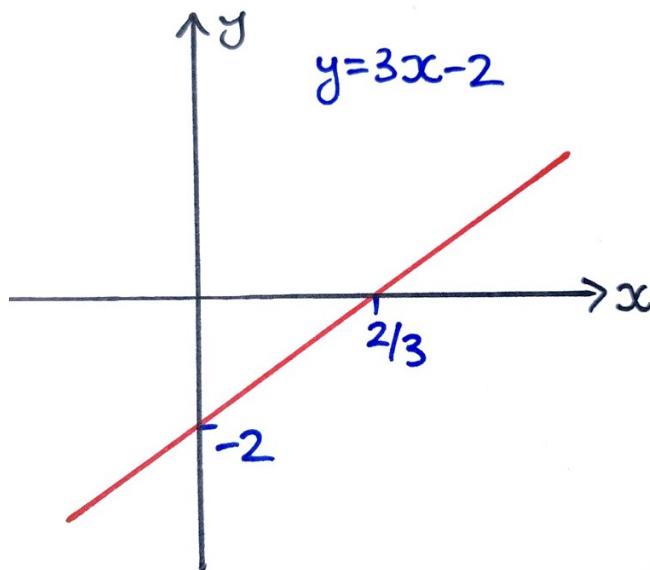
(a) $y = 3x + 2$

y -intercept: $(0, 2)$. Setting $y = 0$ and solving for x : $3x + 2 = 0$ and thus $x = -\frac{2}{3}$.
So we find that the x -intercept is at $\left(-\frac{2}{3}, 0\right)$.



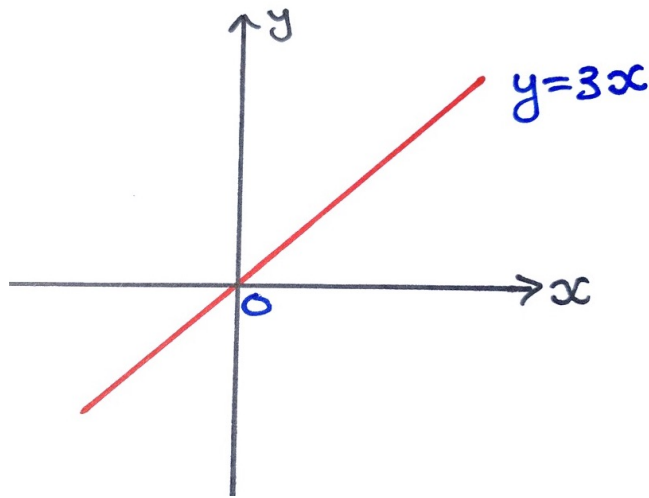
(b) $y = 3x - 2$

y -intercept: $(0, -2)$. Setting $y = 0$ and solving for x , we find that the x -intercept is at $\left(\frac{2}{3}, 0\right)$.



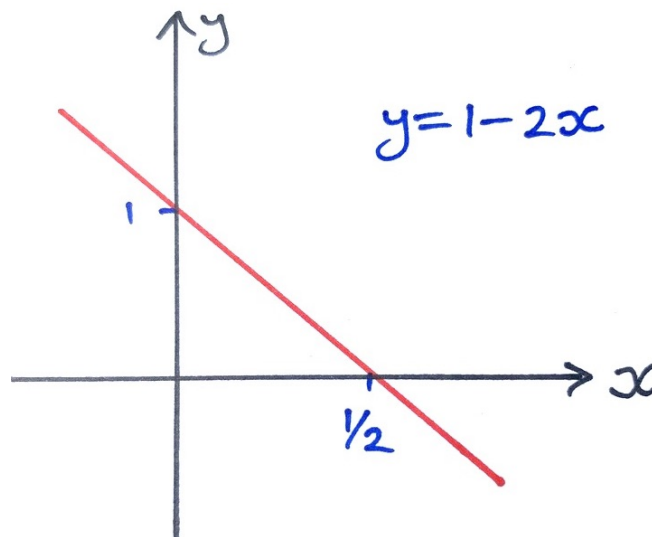
(c) $y = 3x$

Both the x -intercept and the y -intercept are at the origin $(0,0)$, so we could choose any other point. For example, let $x = 1$, then $y = 3(1) = 3$, so the graph passes through $(1,3)$.



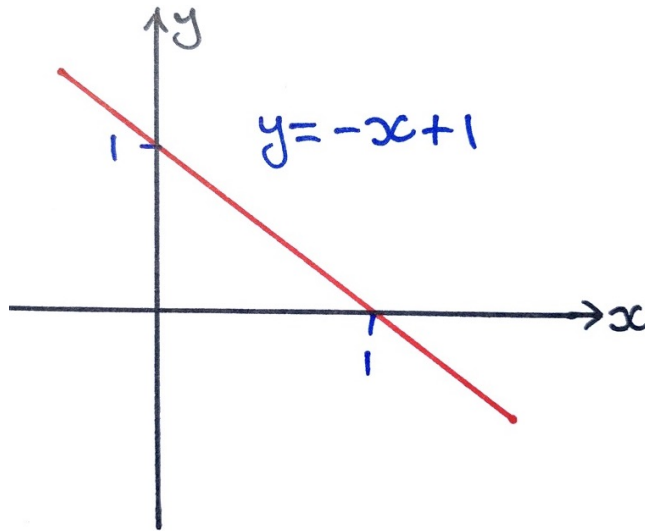
(d) $y = 1 - 2x$

y -intercept: $(0, 1)$. Setting $y = 0$ and solving for x , we find that the x -intercept is at $\left(\frac{1}{2}, 0\right)$.



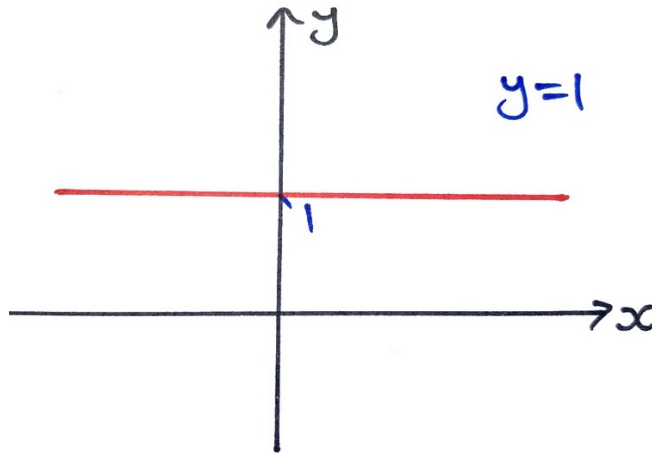
(e) $y = -x + 1$

y -intercept: $(0, 1)$. x -intercept: $(1, 0)$.



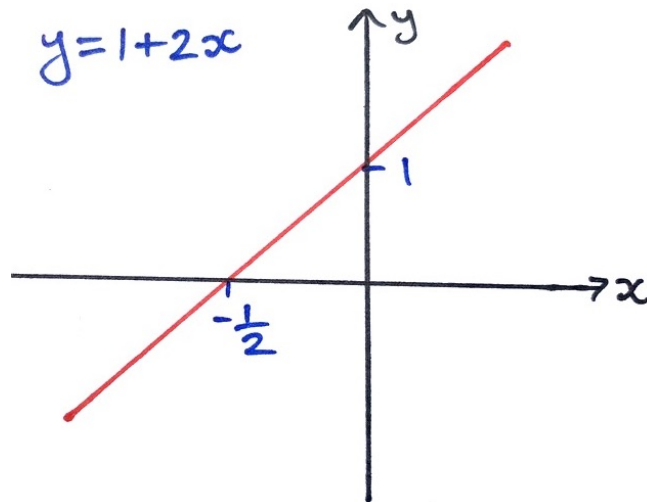
(f) $y = 1$

This is a constant function, so it has zero gradient everywhere and thus is a horizontal line with constant height $y = 1$.



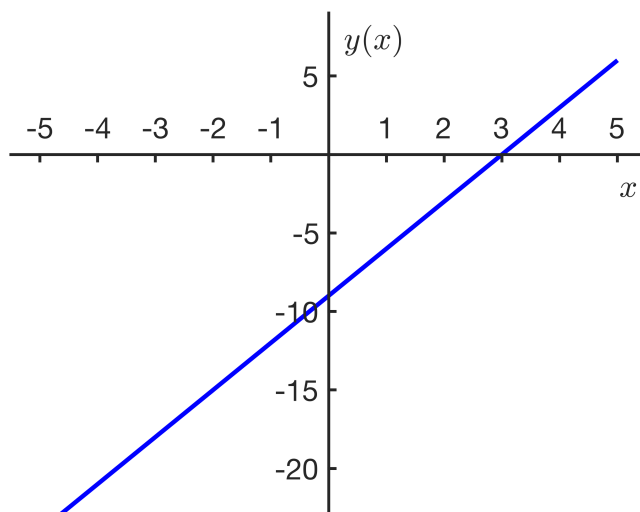
(g) $y = 1 + 2x$

y -intercept: $(0, 1)$. x -intercept: $\left(-\frac{1}{2}, 0\right)$.

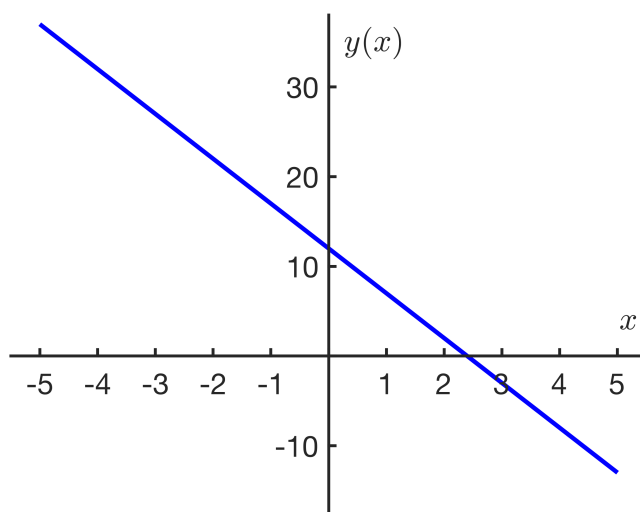


3. Using technology (MATLAB or otherwise), plot the following polynomial functions. You may choose your own range of x , unless otherwise stated.

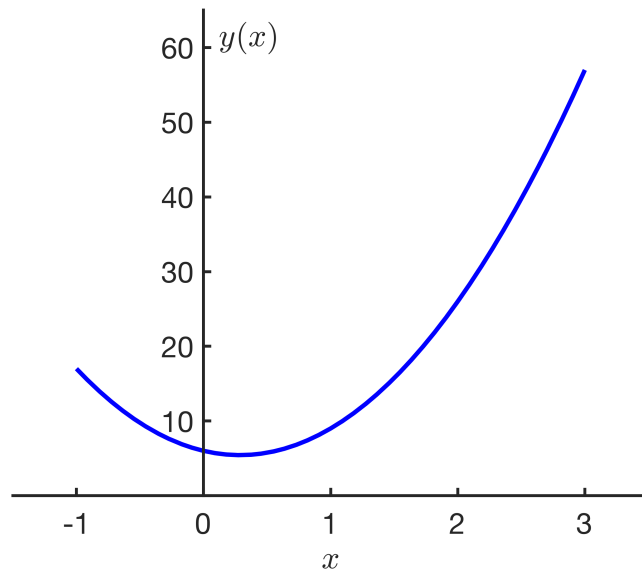
(a) $y = 3x - 9$



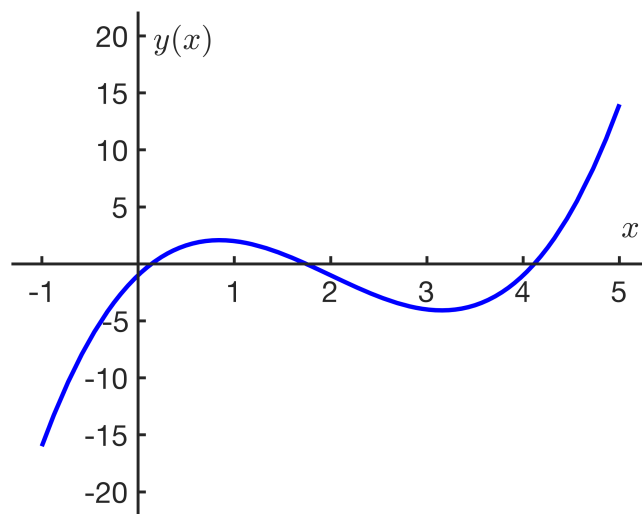
(b) $y = -5x + 12$



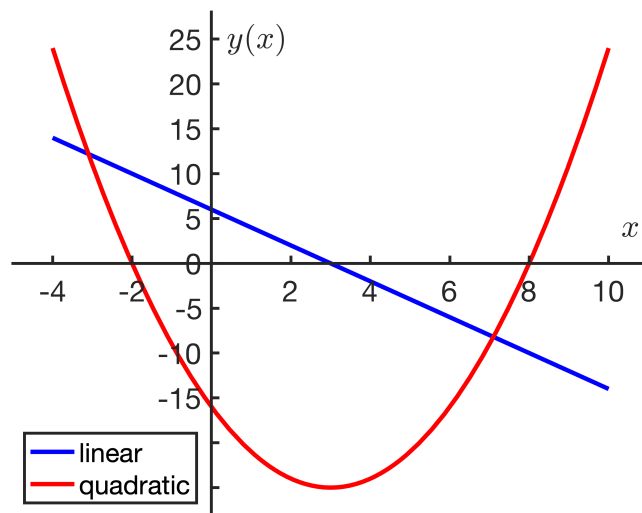
(c) $y = 7x^2 - 4x + 6$ in the range $-1 \leq x \leq 3$



(d) $y = x^3 - 6x^2 + 8x - 1$ in the range $-1 \leq x \leq 5$



- (e) $y = -2x + 6$ and $y = x^2 - 6x - 16$ in the range $-4 \leq x \leq 10$
(Plot both of these functions on the same set of axes)



4. Determine the equation of the straight line that passes through the points:

(a) $(-2, 9)$ and $(1, -6)$

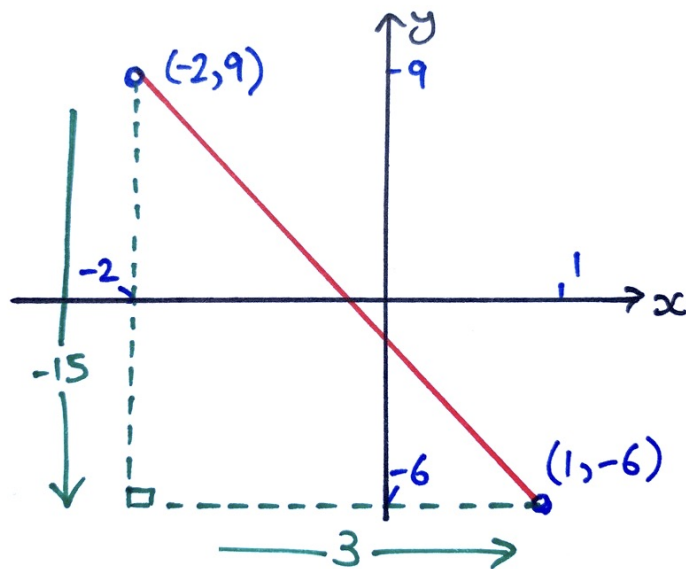
(b) $(-1, -8)$ and $(3, 4)$

(c) $(3, -13)$ and $(5, -17)$

(d) $(1, 5.5)$ and $(6, 3)$

Solution:

(a) $(-2, 9)$ and $(1, -6)$



The equation of a straight line has the form:

$$y = mx + c$$

First, we determine the gradient m :

$$\begin{aligned}
 m &= \frac{\Delta y}{\Delta x} \\
 &= \frac{(-6) - 9}{1 - (-2)} \\
 &= \frac{-15}{3} \\
 &= -5
 \end{aligned}$$

So now the equation is:

$$y = -5x + c$$

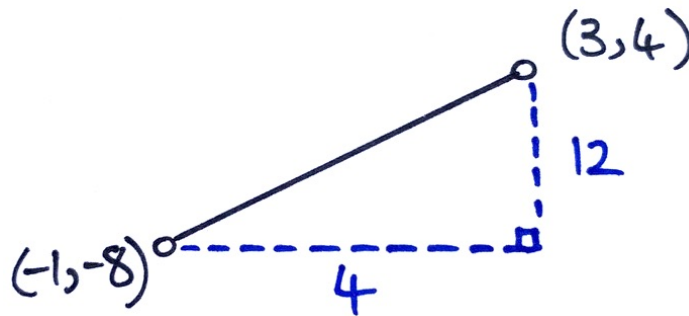
Substitute the co-ordinates of one of the points into this equation to determine c . We will use $(-2, 9)$:

$$\begin{aligned}
 9 &= -5(-2) + c \\
 9 &= 10 + c \\
 \therefore c &= 9 - 10 = -1
 \end{aligned}$$

Hence,

$$y = -5x - 1$$

(b) $(-1, -8)$ and $(3, 4)$



Starting with the equation of the straight line:

$$y = mx + c$$

First, determine the gradient m :

$$m = \frac{\Delta y}{\Delta x} = \frac{4 - (-8)}{3 - (-1)} = \frac{12}{4} = 3$$

So the equation is now:

$$\therefore y = 3x + c$$

Substitute in one of the points that the graph passes through to determine the value of c . We will use $(3, 4)$:

$$4 = 3(3) + c$$

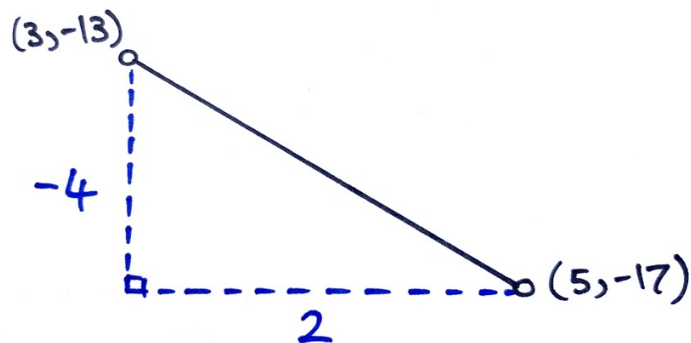
$$4 = 9 + c$$

$$\therefore c = 4 - 9 = -5$$

Hence,

$$y = 3x - 5$$

(c) $(3, -13)$ and $(5, -17)$



Starting with the equation of the straight line:

$$y = mx + c$$

First, determine the gradient m :

$$m = \frac{\Delta y}{\Delta x} = \frac{-17 - (-13)}{5 - 3} = \frac{-4}{2} = -2$$

So the equation becomes:

$$y = -2x + c$$

Substitute in one of the points to determine c . We use $(3, -13)$:

$$-13 = -2(3) + c$$

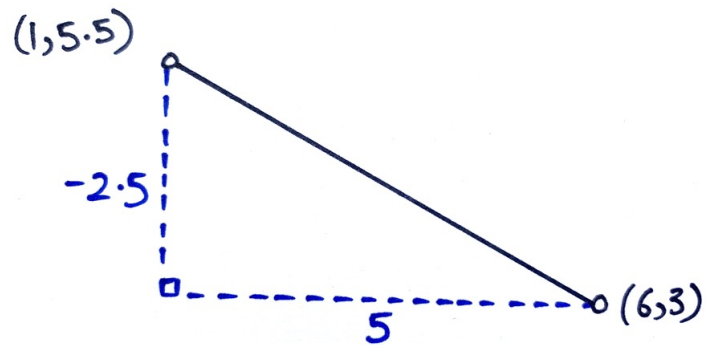
$$-13 = -6 + c$$

$$\therefore c = 6 - 13 = -7$$

Hence,

$$y = -2x - 7$$

(d) $(1, 5.5)$ and $(6, 3)$



Starting with the equation of the straight line:

$$y = mx + c$$

First, determine the gradient m :

$$m = \frac{\Delta y}{\Delta x} = \frac{3 - 5.5}{6 - 1} = \frac{-2.5}{5} = -\frac{1}{2}$$

Hence we have:

$$\therefore y = -\frac{1}{2}x + c$$

Substitute in the co-ordinates of point $(6, 3)$ and solve for c :

$$3 = -\frac{1}{2}(6) + c$$

$$3 = -3 + c$$

$$\therefore c = 3 + 3 = 6$$

Hence,

$$y = -\frac{1}{2}x + 6$$

5. You have been tasked with analysing the behaviour of a complex component in a circuit. You have a dataset of current (I) and voltage (V) measurements, but it is not immediately obvious if the relationship is linear. The data points are as follows:

I (amps)	V (volts)
1	3
2	5
3	8
4	11
5	15

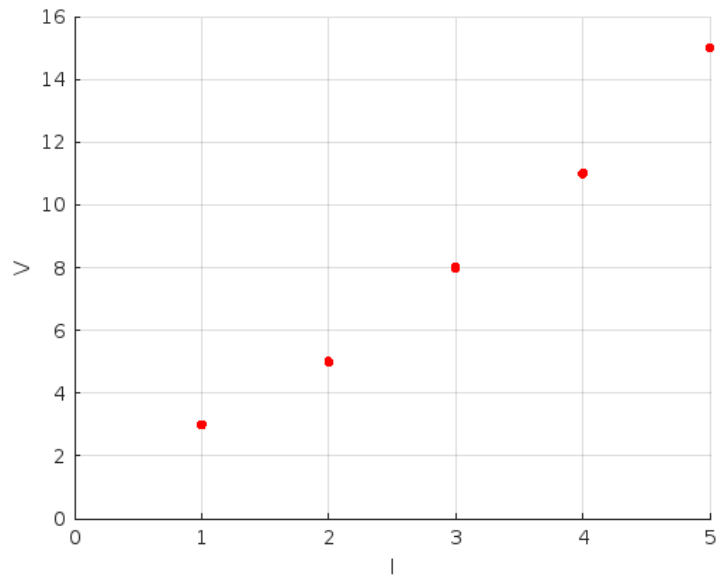
First, determine whether the relationship between current (I) and voltage (V) is linear. If it is, find the linear equation that describes this relationship. Then, predict the voltage (V) when the current is 6. How might understanding this behaviour be critical for designing and optimizing circuits in practical engineering applications?

Solution:

We could tackle this in a number of ways. One way would be to look at the difference in each of the V numbers, given that the increment in I values is fixed at 1 A:

I (amps)	V (volts)	Difference
1	3	n/a
2	5	$5 - 3 = 2$
3	8	$8 - 5 = 3$
4	11	$11 - 8 = 3$
5	15	$15 - 11 = 4$

Another way would be to plot the co-ordinates:



In either case we can establish that the relationship between the current and voltage is non-linear. As such, we cannot express this relationship with a simple linear equation ($V = mI + c$).

Understanding this non-linear behaviour is essential in practical electrical engineering applications because it signifies that the component being analysed does not adhere to Ohm's Law ($V = IR$) for resistance. This non-linearity may be due to various factors such as temperature, material properties, or circuit configuration. Engineers would need to consider these complexities when designing and optimising circuits to ensure accurate and reliable performance.

6. In a market, when there is high demand for a product compared to its supply (before the market gets saturated with product *A* or a similar one from a competitor), we can roughly estimate the relationship between the profit of the company and the number of units manufactured using a linear model.

You are a financial analyst in the employ of the **Persimmon** company, which has just developed a new product called the **uPhone**. Public reception has been extremely positive, and market research suggests that most people are very keen to purchase a **uPhone** as soon as possible. Currently the supply of **uPhones** coming out of the factories is relatively low due to the complex engineering processes and rare metals required in the manufacturing process.

In the first quarter of this year, the company made and sold 1500 units, and reported a profit of £60,000. In the second quarter, the company were able to increase production to 6000 units. All of these were sold and a profit of £960,000 was made.

(a) How many **uPhones** need to be sold each quarter in order to break even during that financial quarter?

(b) In the third quarter, supply managers are warning that production may slow to 5000 units. Assuming that public demand for the product remains high, how much profit do you forecast?

Solution:

The key point to observe here is in the first paragraph: “we can roughly approximate the relationship between the profit . . . and the number of units manufactured with a linear model.” So we are going to attempt to determine a linear (straight line) relationship between these two quantities of *profit* and the *number of units*, bearing in mind the limited circumstances this relationship will apply to.

For each quarter, let the profit of our company, in pounds, be P . And let the number of units manufactured be N . Then we will seek a relationship of the form:

$$P = mN + c$$

and we will first need to determine the gradient m and the intercept c . The other information we are given that can help with this is in the third paragraph: “In the first quarter of this year, the company made and sold 1500 units, and reported a

profit of £60,000. In the second quarter, we were able to increase production to 6000 units. All of these were sold and a profit of £960,000 was made.” So, when $N = 1500$, we have $P = 60000$. And when $N = 6000$, it follows that $P = 960000$. Let’s substitute this information in to our linear equation:

$$\begin{aligned} 60000 &= 1500m + c \\ 960000 &= 6000m + c \end{aligned}$$

We could solve this pair of linear simultaneous equations for m and c in several ways: we could rearrange one of the equations to make c the subject, and then substitute it into the other equation, which would result in an equation only involving m which we could solve. By the end of the module we will learn another method for solving this using matrix algebra. However, since we are dealing with the equation of a straight line we can first find the gradient m :

$$m = \frac{\Delta P}{\Delta N} = \frac{960000 - 60000}{6000 - 1500} = \frac{900000}{4500} = 200$$

So every **uPhone** sold contributes to the overall profit by £200.

Then we have:

$$P = 200N + c$$

And we can substitute either profit-units pair into this and solve for c . For example, when $N = 1500$, we know that $P = 60000$, so:

$$\begin{aligned} 60000 &= 200(1500) + c \\ 60000 &= 300000 + c \\ c &= 60000 - 300000 \\ \therefore c &= -240000 \end{aligned}$$

And hence we have the linear relationship:

$$P = 200N - 240000$$

(a) How many **uPhones** need to be sold each quarter in order to break even during that financial quarter?

Solution:

To break even, we need move from loss into profit, so we are interested in the point at which we clear the threshold of zero profit. For what number of units N does this occur?

Set $P = 0$ and solve for N :

$$0 = 200N - 240000$$

$$200N = 240000$$

$$N = 240000/200$$

$$N = 1200$$

So at exactly $N = 1200$ units manufactured (and sold) we have exactly $P = 200(1200) - 240000 = 0$ profit, and the company breaks even.

(b) In the third quarter, supply managers are warning that production may slow to 5000 units. Assuming that public demand for the product remains high, how much profit do you forecast?

Solution:

The model

$$P = 200N - 240000$$

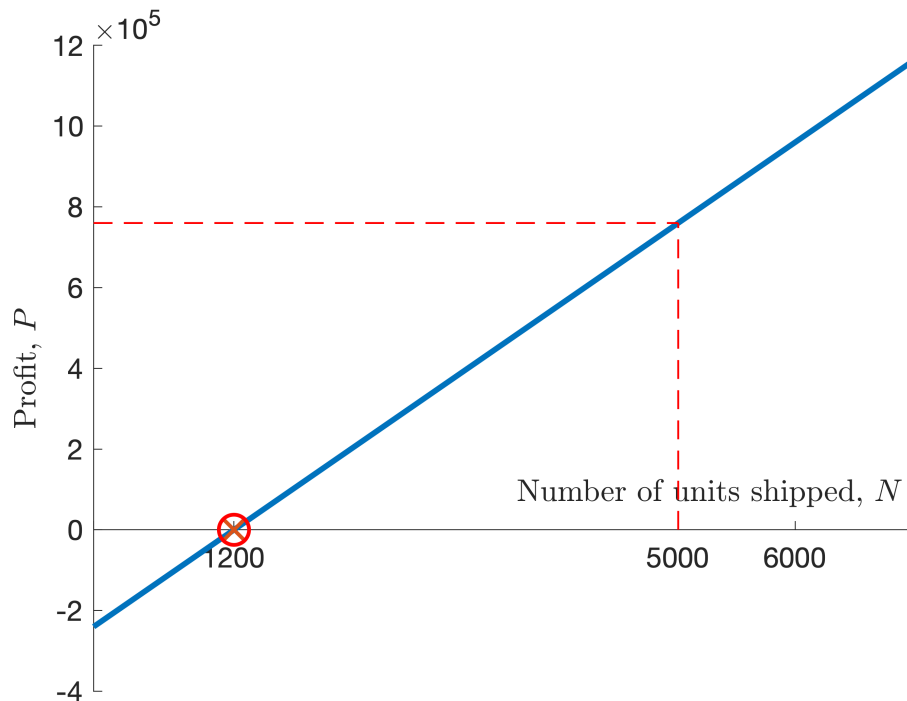
allows us to easily calculate how profit P depends on the number N of units, so simply substitute in $N = 5000$ and evaluate:

$$P = 200(5000) - 240000 = 760000$$

So a profit of £760,000 is anticipated.

Note that this is a reasonable prediction to make because (i) we are told that we are able to assume that the conditions of this model (the high demand) are met, and that (ii) 5000 units per quarter is roughly on the same order as the other quantities that we have been working with. If in the next quarter the company upscaled production to 20 million units we could not be so sure that the model (and hence, the prediction of profit) is reasonable - as many conditions may have changed. Market conditions will certainly change as the uPhone becomes much more widely available and retailer start to offer discounts, and it becomes less of a prestige symbol. There will likely also be a lot more overheads as we will have had to build new factories, employ additional workers, or invest in new processes, additional transportation to markets, perhaps a new mass-market advertising campaign, etc. The point here is that we must be aware of assumptions involved in constructing a model and hence the limitations in reliability when using it to make predictions in situations for which it may not be designed. The supply costs involved in producing several thousand of a product may be very different from those involved in supplying it in the millions in the same time interval, so we would have to be very careful about using a model developed for one set of circumstances to make predictions for the other set of circumstances.

It would also be sensible to sketch or plot the model relationship:



7. Consider the quadratic function $y = x^2 + 3x - 4$

- (a) Where does $y = x^2 + 3x - 4$ cross the y -axis?
- (b) Where does $y = x^2 + 3x - 4$ cross the x -axis?
- (c) Is it $\cup$ -shaped, or $\cap$ -shaped?
- (d) Sketch the graph of this function.

Solution:

- (a) Crosses the y -axis when $x = 0$:

$$\begin{aligned}y &= 0^2 + 3(0) - 4 \\&= -4\end{aligned}$$

So at the y -intercept of -4 .

- (b) The curve crosses the x -axis at $y = 0$:

$$0 = x^2 + 3x - 4$$

Here, $a = 1$, $b = 3$ and $c = -4$, and we can solve this using the quadratic formula:

$$\begin{aligned}x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\&= \frac{-3 \pm \sqrt{3^2 - 4(1)(-4)}}{2 \times 1} \\&= \frac{-3 \pm \sqrt{25}}{2} \\&= \frac{-3 \pm 5}{2}\end{aligned}$$

Hence we have the roots:

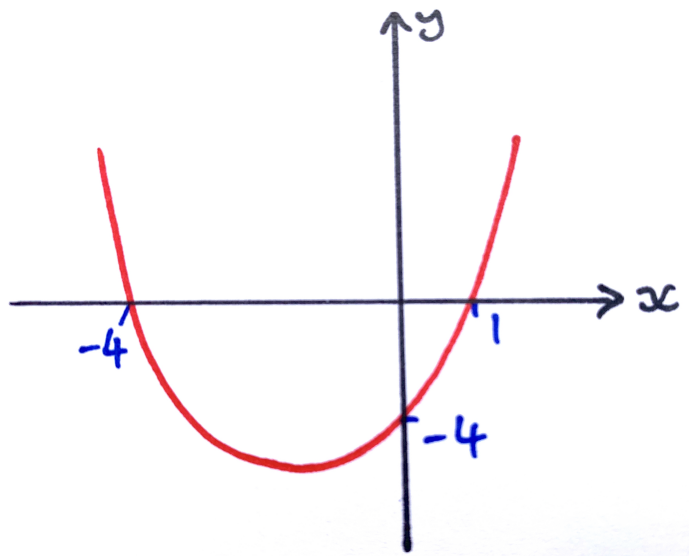
$$x_1 = \frac{-3 - 5}{2} = \frac{-8}{2} = -4$$

and

$$x_2 = \frac{-3 + 5}{2} = \frac{2}{2} = 1$$

(c) This graph will be U-shaped as $a > 0$.

(d) Given the shape of the parabola, and the important points (x and y -intercepts), we can sketch the curve:



8. Repeat the process of Question 7 for the functions:

(i) $y = 16 - x^2$

(ii) $y = 2x^2 + 4x - 20$

Check your results by plotting the curves in MATLAB or other software (such as EXCEL, Desmos or GeoGebra).

Solution:

(i) For $y = 16 - x^2$

(a) Crosses the y -axis when $x = 0$:

$$\begin{aligned} y &= 16 - 0^2 \\ &= 16 \end{aligned}$$

(b) Crosses the x -axis at $y = 0$:

$$0 = 16 - x^2$$

In this case, we could simply re-write it as:

$$x^2 = 16$$

and so taking square roots of both sides, we immediately obtain

$$x = \pm\sqrt{16} = \pm 4$$

Or we could still use the quadratic formula. To do this, write the equation as:

$$-x^2 + 0x + 16 = 0$$

Here, $a = -1$, $b = 0$ and $c = 16$, and thus:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-0 \pm \sqrt{0^2 - 4(-1)(16)}}{2 \times (-1)} \\ &= \frac{\pm \sqrt{64}}{-2} \\ &= \frac{\pm 8}{-2} \end{aligned}$$

Hence we have the roots:

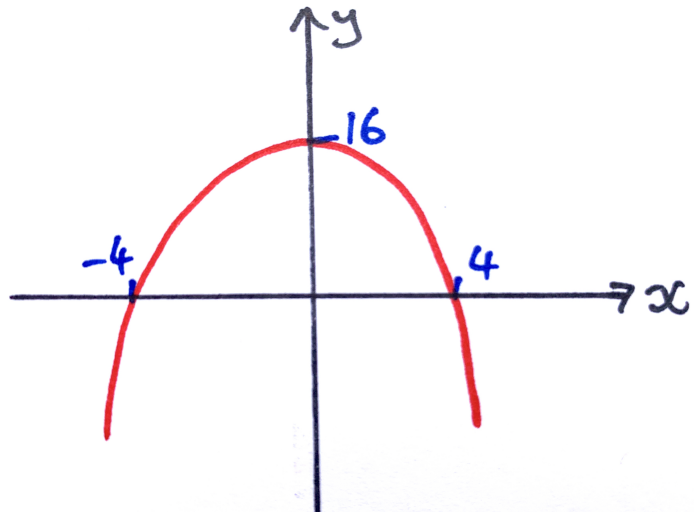
$$x_1 = \frac{-8}{-2} = 4$$

and

$$x_2 = \frac{8}{-2} = -4$$

(c) This graph will be $\cap$ -shaped as $a < 0$.

(d) Given the shape of the parabola, and the important points (x and y -intercepts), we can sketch the curve:



(ii) For $y = 2x^2 + 4x - 20$

(a) Crosses the y -axis when $x = 0$:

$$\begin{aligned} y &= 2(0)^2 + 4(0) - 20 \\ &= -20 \end{aligned}$$

(b) Crosses the x -axis at $y = 0$:

$$0 = 2x^2 + 4x - 20$$

Here, $a = 2$, $b = 4$ and $c = -20$, and applying the quadratic formula:

$$\begin{aligned}
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
&= \frac{-4 \pm \sqrt{4^2 - 4(2)(-20)}}{2 \times 2} \\
&= \frac{-4 \pm \sqrt{176}}{4} \\
&= \frac{-4 \pm 4\sqrt{11}}{4} \\
&= -1 \pm \sqrt{11}
\end{aligned}$$

Hence we have the roots:

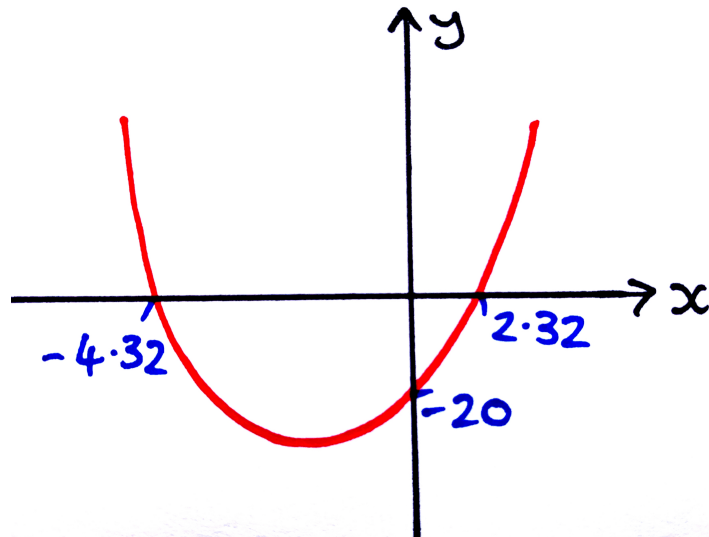
$$x_1 = -1 + \sqrt{11} = 2.32 \dots$$

and

$$x_2 = -1 - \sqrt{11} = -4.32 \dots$$

(c) This graph will be $\cup$ -shaped as $a > 0$.

(d) Given the shape of the parabola, and the important points (x and y -intercepts), we can sketch the curve:



9. Calculate the roots of the following polynomials:

(a) $y = x^2 + 6x - 27$

(b) $y = x^2 + 2x - 48$

(c) $y = 3x^2 + 11x + 8$

(d) $y = 6x^2 - 23x + 20$

Solution:

(a) $y = x^2 + 6x - 27$

Here, the coefficients are $a = 1$, $b = 6$ and $c = -27$. Substituting these into the quadratic formula gives:

$$\begin{aligned}x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\&= \frac{-6 \pm \sqrt{6^2 - 4(1)(-27)}}{2 \times 1} \\&= \frac{-6 \pm \sqrt{144}}{2} \\&= \frac{-6 \pm 12}{2} \\&= -3 \pm 6\end{aligned}$$

Hence, the roots are:

$$x_1 = -3 - 6 = -9$$

and

$$x_2 = -3 + 6 = 3$$

(b) $y = x^2 + 2x - 48$

In this case, the coefficients are $a = 1$, $b = 2$ and $c = -48$. Substituting these into the quadratic formula:

$$\begin{aligned}x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\&= \frac{-2 \pm \sqrt{6^2 - 4(1)(-48)}}{2 \times 1} \\&= \frac{-2 \pm \sqrt{196}}{2} \\&= \frac{-2 \pm 14}{2} \\&= -1 \pm 7\end{aligned}$$

Hence the roots are:

$$x_1 = -1 - 7 = -8$$

and

$$x_2 = -1 + 7 = 6$$

(c) $y = 3x^2 + 11x + 8$

Here, the coefficients are $a = 3$, $b = 11$ and $c = 8$. Substituting these into the quadratic formula gives:

$$\begin{aligned}
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
&= \frac{-11 \pm \sqrt{11^2 - 4(3)(8)}}{2 \times 3} \\
&= \frac{-11 \pm \sqrt{25}}{6} \\
&= \frac{-11 \pm 5}{6}
\end{aligned}$$

Hence the roots are:

$$x_1 = \frac{-11 - 5}{6} = \frac{-16}{6} = \frac{-8}{3}$$

and

$$x_2 = \frac{-11 + 5}{6} = \frac{-6}{6} = -1$$

(d) $y = 6x^2 - 23x + 20$

Here, the coefficients are $a = 6$, $b = -23$ and $c = 20$. Substituting these into the quadratic formula gives:

$$\begin{aligned}
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
&= \frac{-(-23) \pm \sqrt{(-23)^2 - 4(6)(20)}}{2 \times 6} \\
&= \frac{23 \pm \sqrt{49}}{12} \\
&= \frac{23 \pm 7}{12}
\end{aligned}$$

Hence the roots are:

$$x_1 = \frac{23 - 7}{12} = \frac{16}{12} = \frac{4}{3}$$

and

$$x_2 = \frac{23 + 7}{12} = \frac{30}{12} = \frac{5}{2}$$

10. If the roots of the quadratic $y = ax^2 + bx + c$ are x_1 and x_2 then it is possible to rewrite the quadratic in its factorised form as follows:

$$y = a(x - x_1)(x - x_2)$$

(a) Use this information to rewrite the four quadratics that appear in Question 9.

(b) Expand the brackets of your new form in each case to demonstrate that it is equivalent to the original form.

Solution:

(a)

For $x^2 + 6x - 27$:

Here, we found the roots to be $x_1 = -9$ and $x_2 = 3$.

Given that $a = 1$, we can substitute into the factorised form as follows:

$$\begin{aligned} y &= a(x - x_1)(x - x_2) \\ &= 1(x - -9)(x - 3) \end{aligned}$$

hence, the factorised form is:

$$y = (x + 9)(x - 3).$$

For $x^2 + 2x - 48$:

Here, we found the roots to be $x_1 = -8$ and $x_2 = 6$.

Given that $a = 1$, we can substitute into the factorised form as follows:

$$\begin{aligned} y &= a(x - x_1)(x - x_2) \\ &= 1(x - -8)(x - 6) \end{aligned}$$

hence, the factorised form is:

$$y = (x + 8)(x - 6).$$

For $3x^2 + 11x + 8$:

Here, we found the roots to be $x_1 = -\frac{8}{3}$ and $x_2 = -1$.

Given that $a = 3$, we can substitute into the factorised form as follows:

$$\begin{aligned} y &= a(x - x_1)(x - x_2) \\ &= 3 \left(x - -\frac{8}{3} \right) \left(x - -1 \right) \\ &= (3x - -8)(x - -1) \end{aligned}$$

hence, the factorised form is:

$$y = (3x + 8)(x + 1).$$

For $6x^2 - 23x + 20$:

Here, we found the roots to be $x_1 = \frac{4}{3}$ and $x_2 = \frac{5}{2}$.

Given that $a = 6$, we can substitute into the factorised form as follows:

$$\begin{aligned} y &= a(x - x_1)(x - x_2) \\ &= 6 \left(x - \frac{4}{3} \right) \left(x - \frac{5}{2} \right) \quad [\text{splitting the 6 into factors}] \\ &= 3 \left(x - \frac{4}{3} \right) \times 2 \left(x - \frac{5}{2} \right) \end{aligned}$$

hence, the factorised form is:

$$y = (3x - 4)(2x - 5).$$

(b) For parts (a) to (d) multiply out the brackets to confirm the original forms from question 11.

11. Solve $5x^2 + 2x = 4$, giving the solutions correct to 3 significant figures.

Solution:

Firstly rearrange the equation so that it is equal to zero:

$$5x^2 + 2x - 4 = 0.$$

Here, the coefficients are $a = 5$, $b = 2$ and $c = -4$. Substituting these into the quadratic formula gives:

$$\begin{aligned}
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
&= \frac{-2 \pm \sqrt{2^2 - 4(5)(-4)}}{2 \times 5} \\
&= \frac{-2 \pm \sqrt{84}}{10}.
\end{aligned}$$

Hence, the roots are:

$$x_1 = \frac{-2 - \sqrt{84}}{10} = -1.12$$

and

$$x_2 = \frac{-2 + \sqrt{84}}{10} = 0.717$$

12. Solve $x^2 = 4(x - 3)^2$

Solution:

Firstly multiply out the brackets:

$$\begin{aligned}
x^2 &= 4(x - 3)(x - 3) \\
x^2 &= 4(x^2 - 6x + 9) \\
x^2 &= 4x^2 - 24x + 36
\end{aligned}$$

Rearranging all terms to the left-hand side gives $-3x^2 + 24x - 36 = 0$.

Here, the coefficients are $a = -3$, $b = 24$ and $c = -36$. Substituting these into the quadratic formula gives:

$$\begin{aligned}
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
&= \frac{-24 \pm \sqrt{24^2 - 4(-3)(-36)}}{2 \times -3} \\
&= \frac{-24 \pm \sqrt{144}}{-6} \\
&= \frac{-24 \pm 12}{-6}.
\end{aligned}$$

Hence, the roots are:

$$x_1 = \frac{-24 - 12}{-6} = \frac{-36}{-6} = 6$$

and

$$x_2 = \frac{-24 + 12}{-6} = \frac{-12}{-6} = 2$$

13. Solve $\frac{5}{x+2} + \frac{9}{x-2} = 2$

Solution: Note that there is more than one way to solve this.

Firstly multiply each term by $x+2$:

$$\begin{aligned}\frac{5}{x+2} + \frac{9}{x-2} &= 2 \\ \frac{5(x+2)}{x+2} + \frac{9(x+2)}{x-2} &= 2(x+2) \\ 5 + \frac{9(x+2)}{x-2} &= 2(x+2).\end{aligned}$$

Now multiply each term by $x-2$:

$$\begin{aligned}5 + \frac{9(x+2)}{x-2} &= 2(x+2) \\ 5(x-2) + \frac{9(x+2)(x-2)}{x-2} &= 2(x+2)(x-2) \\ 5(x-2) + 9(x+2) &= 2(x+2)(x-2).\end{aligned}$$

Multiplying out and rearranging gives:

$$\begin{aligned}5(x-2) + 9(x+2) &= 2(x+2)(x-2) \\ 5x - 10 + 9x + 18 &= 2(x^2 - 4) \\ 14x + 8 &= 2x^2 - 8 \\ -2x^2 + 14x + 16 &= 0.\end{aligned}$$

Here, the coefficients are $a = -2$, $b = 14$ and $c = 16$. Substituting these into the quadratic formula gives:

$$\begin{aligned}
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
&= \frac{-14 \pm \sqrt{14^2 - 4(-2)(16)}}{2 \times -2} \\
&= \frac{-14 \pm \sqrt{324}}{-4} \\
&= \frac{-14 \pm 18}{-4}.
\end{aligned}$$

Hence, the roots are:

$$x_1 = \frac{-14 - 18}{-4} = \frac{-32}{-4} = 8$$

and

$$x_2 = \frac{-14 + 18}{-4} = \frac{4}{-4} = -1$$

14. The quadratic formula is being used to solve a quadratic equation. After substitution of the coefficients into the formula, we have

$$x = \frac{-7 \pm \sqrt{17}}{4}$$

Work out the original quadratic equation, giving your answer in the form $ax^2 + bx + c = 0$.

Solution:

The quadratic formula is in the form:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Therefore $-b = -7$, hence $b = 7$. From the denominator we can deduce that $2a = 4$. Therefore $a = 2$. Finally we can determine the value of c from the discriminant:

$$\begin{aligned} b^2 - 4ac &= 17 \\ 7^2 - 4 \times 2 \times c &= 17 \\ 49 - 8c &= 17 \\ -8c &= -32 \\ c &= 4 \end{aligned}$$

Therefore the quadratic equation is $2x^2 + 7x + 4 = 0$.

15. Solve the equation $\frac{2^{(n^2)}}{2^n \times 2^6} = 1$

Solution:

Before we can use the quadratic equation, the base 2 must be eliminated. Let's use the rules of indices and logs to simplify the equation:

$$\frac{2^{(n^2)}}{2^n \times 2^6} = 1 \quad [\text{add the powers on the denominator}]$$

$$\frac{2^{(n^2)}}{2^{n+6}} = 1 \quad [\text{subtract the powers}]$$

$$2^{(n^2-n-6)} = 1 \quad [\text{take } \log_2 \text{ on both sides}]$$

$$n^2 - n - 6 = \log_2(1)$$

$$n^2 - n - 6 = 0.$$

This is now a quadratic equation where $a = 1$, $b = -1$ and $c = -6$. Substituting these into the quadratic formula gives:

$$\begin{aligned} n &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-6)}}{2 \times 1} \\ &= \frac{1 \pm \sqrt{25}}{2} \\ &= \frac{1 \pm 5}{2}. \end{aligned}$$

Hence, the solutions are:

$$n_1 = \frac{1 - 5}{2} = \frac{-4}{2} = -2$$

and

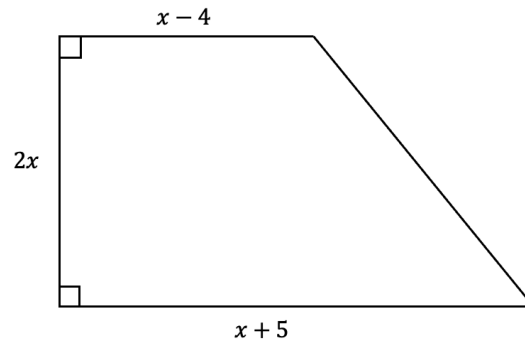
$$n_2 = \frac{1 + 5}{2} = \frac{6}{2} = 3$$

16. Shown in the diagram below is a trapezium (not drawn to scale) where all measurements are in centimetres.

The area of the trapezium is 351 m^2 . Determine the value of x .

Solution:

The area of a trapezium is $A = \frac{h}{2}(a + b)$, where a and b are the lengths and h is the vertical height. Substituting our values into the formula gives:



$$A = \frac{h}{2}(a + b)$$

$$351 = \frac{2x}{2}((x-4) + (x+5))$$

$$351 = \frac{2x}{2}(2x+1) \quad [\text{cancelling the 2s}]$$

$$351 = x(2x+1)$$

$$351 = 2x^2 + x$$

$$0 = 2x^2 + x - 351.$$

This is now a quadratic equation where $a = 2$, $b = 1$ and $c = -351$. Substituting these into the quadratic formula gives:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-1 \pm \sqrt{1^2 - 4(2)(-351)}}{2 \times 2} \\ &= \frac{-1 \pm \sqrt{2809}}{4} \\ &= \frac{-1 \pm 53}{4}. \end{aligned}$$

Hence, the solutions are:

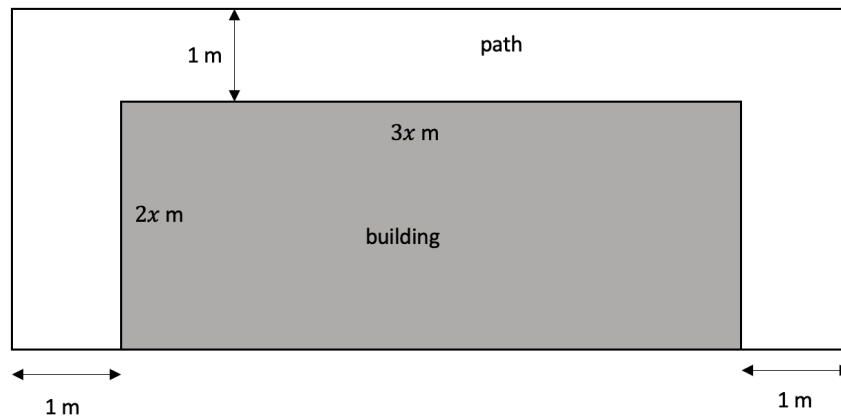
$$x_1 = \frac{-1 - 53}{4} = \frac{-54}{4} = -13.5$$

and

$$x_2 = \frac{-1 + 53}{4} = \frac{52}{4} = 13$$

The only sensible answer is $x = 13$ m, as this will result in positive values for the lengths and height of the trapezium.

17. A rectangular building has a length of $3x$ metres and a width of $2x$ metres. The building has a path of width 1 metre on three of its sides, as depicted in the diagram below. Note that the diagram is not drawn to scale.



Given that the total area of the building and path is 100 m^2 calculate the area of the building.

Solution:

The value of x must be determined before we can calculate the area of the building. The area of the building plus the path is known and is rectangular. The area of a rectangle is $A = lw$, where l is the length and w is the width. Substituting our values into the formula gives:

$$A = lw$$

$$100 = (1 + 3x + 1) \times (2x + 1)$$

$$100 = (3x + 2) \times (2x + 1)$$

$$100 = 6x^2 + 7x + 2$$

$$0 = 6x^2 + 7x - 98$$

This is now a quadratic equation where $a = 6$, $b = 7$ and $c = -98$. Substituting these into the quadratic formula gives:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-7 \pm \sqrt{7^2 - 4(6)(-98)}}{2 \times 6} \\ &= \frac{-7 \pm \sqrt{2401}}{12} \\ &= \frac{-7 \pm 49}{12}. \end{aligned}$$

Hence, the solutions are:

$$x_1 = \frac{-7 - 49}{12} = \frac{-56}{12} = -\frac{14}{3}$$

and

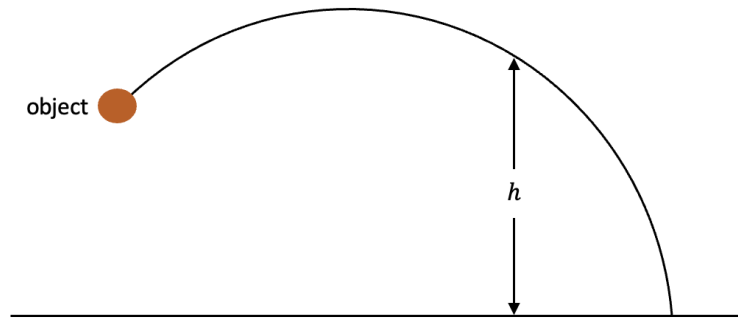
$$x_2 = \frac{-7 + 49}{12} = \frac{42}{12} = 3.5.$$

The only sensible answer is $x = 3.5$ m, as this will result in positive values for the length and width of the building. Now the value of x is known, we are able to calculate the area of the building:

$$\begin{aligned}
 A &= lw \\
 &= 3x \times 2x \\
 &= 10.5 \times 7 \\
 &= 73.5
 \end{aligned}$$

Therefore the area of the building is 73.5 m^2 .

18. An object is being launched from an initial height of 2 m above the ground. The object follows a parabola in the form $h = 2 + 6t - 5t^2$, where h is the height in metres above the ground t seconds after it has been launched.



The object hits the ground after T seconds. Calculate the value of T , giving the answer to 2 decimal places.

Solution:

When the object hits the ground the value of h must be zero. Therefore we have $0 = 2 + 6T - 5T^2$. This is a quadratic equation where $a = -5$, $b = 6$ and $c = 2$. Substituting these into the quadratic formula gives:

$$\begin{aligned}
T &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
&= \frac{-6 \pm \sqrt{6^2 - 4(-5)(2)}}{2 \times -5} \\
&= \frac{-6 \pm \sqrt{76}}{-10}.
\end{aligned}$$

Hence, the solutions are:

$$T_1 = \frac{-6 - \sqrt{76}}{-10} = 1.47$$

and

$$T_2 = \frac{-6 + \sqrt{76}}{-10} = -0.27.$$

The only sensible answer is $T = 1.47$ s, as this is the only result that is positive. Furthermore, even if both times were positive, the largest value of T would be chosen as the first, whilst a correct result for the solution of the parabola, is not correct for the problem, as the object is thrown from a height of 2 m.

19. Access the **Interactive Function Explorer** on the module Blackboard site. Navigate to the “logarithmic” tab, featuring the general logarithmic equation:

$$y = A \ln(Bx) + C$$

Determine how the two parameters affect the shape of the graph:

- (a) A
- (b) C

What happens when you try to calculate $\ln(0)$ or $\log$ of a negative number?

Solution:

(a) Increasing A stretches the curve in the vertical direction.

(b) C shifts the curve up/down.

Logs are not defined for zero and non-positive inputs. i.e. $\log$ of zero or any negative number does not exist.

20. Access the **Interactive Function Explorer** on the module Blackboard site. Navigate to the “exponential” tab, featuring the general exponential function:

$$y = A e^{Bx} + C$$

Determine how the two parameters affect the shape of the graph:

(a) A

(a) B

(b) C

What conditions must be satisfied by B in order to obtain exponential growth, or to obtain exponential decay?

Solution:

(a) Increasing the magnitude of A stretches the curve in the vertical axis.

(b) B controls the rate of growth/decay. If $B > 0$ then the curve exhibits exponential growth, while if $B < 0$ then the curve exhibits exponential decay.

(c) C shifts the curve up/down without changing the shape.

21. Solve the following exponential equations for x :

(a) $e^x = 9.5$

(b) $5e^x = 50.6$

$$(c) \ 27.3 e^x - 12 = 112.7$$

$$(d) \ -14.6 e^x + 7.5 = 4.2$$

$$(e) \ 29 - 6.3 e^x = 15.4$$

$$(f) \ 34.7 e^{4.2x} + 4.3 = 25.1$$

$$(g) \ 117.1 e^{-0.4x} - 15.7 = -3.8$$

Solutions:

$$(a) \ e^x = 9.5$$

Taking the natural log of both sides:

$$\ln(e^x) = \ln(9.5)$$

$$\therefore x = \ln(9.5)$$

$$= 2.25$$

$$(b) \ 5 e^x = 50.6$$

First divide both sides by 5 so that the entire LHS is purely an exponential function:

$$e^x = \frac{50.6}{5}$$

$$\therefore e^x = \frac{253}{25}$$

Then taking the natural log of both sides:

$$x = \ln\left(\frac{253}{25}\right)$$

$$= 2.31$$

$$(c) \ 27.3 e^x - 12 = 112.7$$

Begin by isolating the exponential function on the LHS:

$$\begin{aligned} 27.3 e^x &= 112.7 + 12 \\ \therefore 27.3 e^x &= 124.7 \\ \therefore e^x &= \frac{124.7}{27.3} \end{aligned}$$

Then taking $\ln$ of both sides:

$$\begin{aligned} x &= \ln \left(\frac{1247}{273} \right) \\ &= 1.52 \end{aligned}$$

$$(d) \ -14.6 e^x + 7.5 = 4.2$$

Isolating the exponential function on the LHS:

$$\begin{aligned} -14.6 e^x &= 4.2 - 7.5 \\ \therefore -14.6 e^x &= -\frac{33}{10} \\ \therefore e^x &= \frac{33}{146} \end{aligned}$$

Then taking the natural log of both sides:

$$\begin{aligned}
 x &= \ln \left(\frac{33}{146} \right) \\
 &= -1.49
 \end{aligned}$$

(e) $29 - 6.3 e^x = 15.4$

Isolating the exponential function:

$$\begin{aligned}
 -6.3 e^x &= 15.4 - 29 \\
 \therefore -6.3 e^x &= -\frac{68}{5} \\
 \therefore e^x &= \frac{136}{63}
 \end{aligned}$$

Then taking $\ln$ of both sides:

$$\begin{aligned}
 x &= \ln \left(\frac{136}{63} \right) \\
 &= 0.77
 \end{aligned}$$

(f) $34.7 e^{4.2x} + 4.3 = 25.1$

Isolating the exponential function:

$$\begin{aligned}
 34.7 e^{4.2x} &= 25.1 - 4.3 \\
 \therefore 34.7 e^{4.2x} &= \frac{104}{5} \\
 \therefore e^{4.2x} &= \frac{208}{347}
 \end{aligned}$$

Then taking logs of both sides:

$$4.2x = \ln \left(\frac{208}{347} \right)$$

$$\begin{aligned}\therefore x &= \frac{1}{4.2} \ln \left(\frac{208}{347} \right) \\ &= -0.12\end{aligned}$$

(g) $117.1 e^{-0.4x} - 15.7 = -3.8$

First, isolating the exponential function on the LHS:

$$\begin{aligned}117.1 e^{-0.4x} &= -3.8 + 15.7 \\ \therefore 117.1 e^{-0.4x} &= \frac{119}{10} \\ \therefore e^{-0.4x} &= \frac{119}{1171}\end{aligned}$$

Then taking the natural log of both sides:

$$\begin{aligned}-0.4x &= \ln \left(\frac{119}{1171} \right) \\ \therefore x &= \frac{1}{-0.4} \ln \left(\frac{119}{1171} \right) \\ &= 5.72\end{aligned}$$

22. Solve the following logarithmic equations for x :

(a) $\ln(x) = 0.6$

(b) $15 - \ln(2x) = 10.3$

(c) $2.8 = \log_{10}(0.3x + 6)$

Solutions:

(a) $\ln(x) = 0.6$

As the desired variable x is contained within a natural log function, we can extract it by taking an exponential function of both sides:

$$\begin{aligned} e^{\ln(x)} &= e^{0.6} \\ x &= e^{0.6} \\ &= 1.82 \quad (2 \text{ d.p.}) \end{aligned}$$

(b) $15 - \ln(2x) = 10.3$

First, isolate the natural log term:

$$\begin{aligned} \ln(2x) &= 15 - 10.3 \\ \ln(2x) &= 4.7 \end{aligned}$$

Then taking an exponential function of both sides:

$$\begin{aligned} e^{\ln(2x)} &= e^{4.7} \\ 2x &= e^{4.7} \\ x &= \frac{1}{2} e^{4.7} \\ &= 54.97 \quad (2 \text{ d.p.}) \end{aligned}$$

(c) $2.8 = \log_{10}(0.3x + 6)$

In this example, one side consists of a log with base 10, rather than the natural log which has base e. To invert this and extract the contents, we take both sides as powers of 10 rather than e:

$$10^{2.8} = 10^{\log_{10}(0.3x+6)}$$

$$10^{2.8} = 0.3x + 6$$

Then transposing for x as usual:

$$0.3x = 10^{2.8} - 6$$

$$\begin{aligned} x &= \frac{1}{0.3}(10^{2.8} - 6) \\ &= 2083.19 \quad (2 \text{ d.p.}) \end{aligned}$$

23. The pressure p pascals at height h metres above ground level is given by:

$$p = p_0 e^{-\frac{h}{C}}$$

where p_0 is the pressure at ground level and C is a constant.

Find the pressure p when $p_0 = 1.012 \times 10^5$ Pa, height is $h = 1420$ m and $C = 71500$.

Solution:

As we already have a formula for pressure, simply substitute in the constants and the specific height, and evaluate:

$$\begin{aligned}
 p &= p_0 e^{-\frac{h}{C}} \\
 &= 1.012 \times 10^5 \times e^{-1420/71500} \\
 &= 99209.98 \text{ Pa}
 \end{aligned}$$

For physical problems such as these, be sure to include appropriate units, and rounding if necessary.

24. The current i flowing in a capacitor at time t is given by:

$$i = 12.5 \left(1 - e^{-\frac{t}{CR}} \right)$$

where $R = 30$ kilohms, and the capacitance C is 20 micro-farads.

Determine: (a) the current flowing after 0.5 seconds. (b) the time for the current to reach 10 amperes.

Solution:

(a) Substitute in the parameters and $t = 0.5$, and evaluate the formula we have for current i :

$$\begin{aligned}
 i &= 12.5 \left(1 - e^{-\frac{t}{CR}} \right) \\
 &= 12.5 \left(1 - e^{\frac{-0.5}{20 \times 10^{-6} \times 30 \times 10^3}} \right) \\
 &= 12.5 \left(1 - e^{\frac{-0.5}{6 \times 10^{-1}}} \right) \\
 &= 12.5 \left(1 - e^{-\frac{5}{6}} \right) \\
 &= 7.07 \text{ amperes}
 \end{aligned}$$

(b) Substitute in the parameters and $i = 10$, and simplify first:

$$10 = 12.5 \left(1 - e^{-\frac{t}{20 \times 10^{-6} \times 30 \times 10^3}} \right)$$

$$10 = 12.5 \left(1 - e^{-\frac{t}{0.6}} \right)$$

Now re-arrange the formula to make t the subject:

$$\frac{10}{12.5} = 1 - e^{-\frac{t}{0.6}}$$

$$\frac{4}{5} - 1 = -e^{-\frac{t}{0.6}}$$

$$-\frac{1}{5} = -e^{-\frac{t}{0.6}}$$

$$e^{-\frac{t}{0.6}} = \frac{1}{5}$$

Then taking the natural log of both sides:

$$-\frac{t}{0.6} = \ln \left(\frac{1}{5} \right)$$

$$t = -0.6 \ln \left(\frac{1}{5} \right)$$

$$= 0.97 \text{ seconds (2 d.p.)}$$

25. The resistance R of an electrical conductor at temperature $\theta^\circ C$ is given by $R = R_0 e^{\alpha\theta}$, where α is a constant and $R_0 = 5 \times 10^3$ ohms.

Determine the value of α , correct to 4 significant figures, when $R = 6 \times 10^3$ ohms and $\theta = 1500^\circ C$.

Also, find the temperature, correct to the nearest degree, when the resistance R is 5.4×10^3 ohms.

Solution:

First, determining the value of α when $R = 6 \times 10^3$. Substitute in the values of R , R_0 and θ and transpose for α using logs:

$$R = R_0 e^{\alpha\theta}$$

$$6 \times 10^3 = 5 \times 10^3 \times e^{1500\alpha}$$

$$\frac{6}{5} = e^{1500\alpha}$$

$$1500\alpha = \ln \left(\frac{6}{5} \right)$$

$$\begin{aligned}\alpha &= \frac{1}{1500} \ln \left(\frac{6}{5} \right) \\ &= 1.215 \times 10^{-4} \quad (4 \text{ s.f.})\end{aligned}$$

Now to find θ when $R = 5.4 \times 10^3$:

$$5.4 \times 10^3 = 5 \times 10^3 \times e^{1.215 \times 10^{-4} \theta}$$

$$\frac{27}{25} = e^{1.215 \times 10^{-4} \theta}$$

$$\ln \left(\frac{27}{25} \right) = 1.215 \times 10^{-4} \theta$$

$$\begin{aligned} \theta &= \frac{1}{1.215 \times 10^{-4}} \ln \left(\frac{27}{25} \right) \\ &= 633^\circ\text{C} \quad (\text{nearest degree}) \end{aligned}$$

26. A research group models the population P (in millions) of a particular country by following formula:

$$P = A e^{kt},$$

where t is the time (in years) since 1980, the value of k is 0.0241, and A is a constant that is yet to be determined.

In the year 2000, the population of the country was recorded as 11 million.

- What is the population projected to be in 2020?
- When is the population forecast to exceed 25 million?
- Plot the projected population in MATLAB between the years 1980 and 2050. From the graph, describe the behaviour of this exponential function.
- From the graph, determine when the population is equal to 27 million. Indicate clearly on the graph how you obtain your solution.
- Your colleague uses this formula to predict the population of the country in the year 2500. Discuss the reliability of this prediction.

Solutions:

First, substitute in $P(t = 20) = 11$ and solve to determine the value of A .

$$11 = A e^{20 \times 0.0241}$$

$$11 = A e^{0.4820}$$

$$A = \frac{11}{e^{0.482}} = 6.7930$$

(a) In 2020, $t = 40$. Then:

$$P(t = 40) = 6.7930 e^{40 \times 0.0241} = 17.8124$$

So the population is projected to be 17.8 million in 2020.

(b) Solve for t when $P = 25$:

$$25 = 6.7930 e^{0.0241t}$$

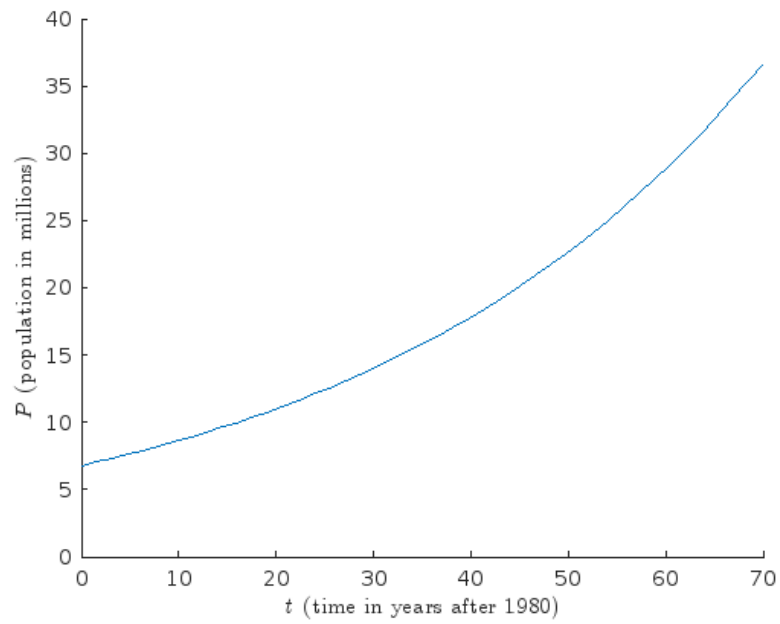
$$e^{0.0241t} = \frac{25}{6.7930} = 3.6803$$

$$0.0241t = \ln(3.6803) = 1.3030$$

$$t = \frac{1.3030}{0.0241} = 54.0657$$

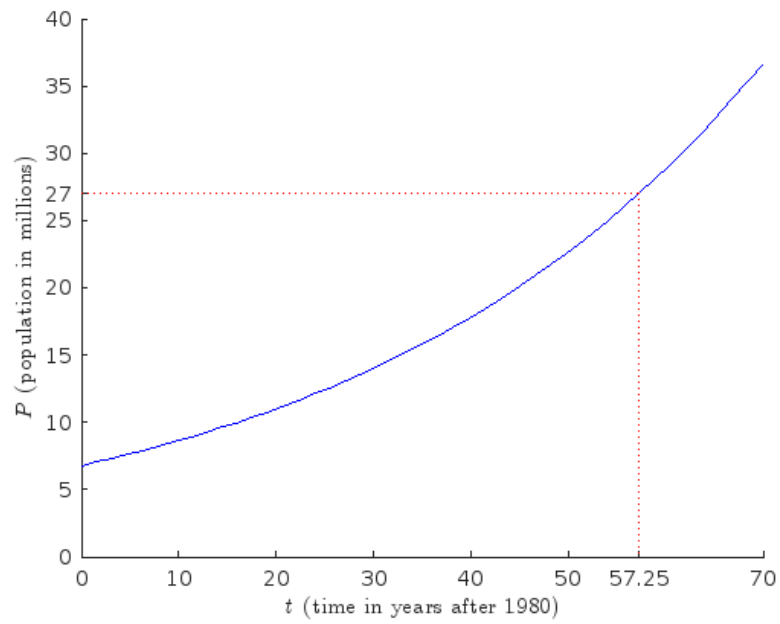
So 54 years after 1980, hence the year is 2034.

(c) Graph:



Grows with increasing rapidity and without bound.

(d) Graph:



Thus, 57 years after 1980, so in 2037.

(e) This is a much greater time frame than the model is calibrated for. Models are not a guaranteed prediction, they are based on the assumption that current circumstances and conditions will never change. Of course, it is very likely that if this pattern were to continue, eventually limitations such as population density, resource, climate and disease that are not accounted for in the current model could become significant.

27. The temperatures θ_1 and θ_2 of a pipe with inner radius r_1 and outer radius r_2 are given by:

$$\theta_1 = -\frac{Q}{2\pi kL} \ln(r_1) \quad \text{and} \quad \theta_2 = -\frac{Q}{2\pi kL} \ln(r_2)$$

where Q is the heat transfer rate, L is the length of the pipe and k is the thermal conductivity. Show that:

$$Q = \frac{2\pi kL(\theta_1 - \theta_2)}{\ln\left(\frac{r_2}{r_1}\right)}$$

Solution:

Combining our expressions for θ_1 and θ_2 :

$$\begin{aligned}
\theta_1 - \theta_2 &= -\frac{Q}{2\pi kL} \ln(r_1) - \left(-\frac{Q}{2\pi kL} \ln(r_2) \right) \\
&= -\frac{Q}{2\pi kL} (\ln(r_1) - \ln(r_2)) \\
&= -\frac{Q}{2\pi kL} (- (\ln(r_2) - \ln(r_1))) \\
&= -\frac{Q}{2\pi kL} \left(- \ln \left(\frac{r_2}{r_1} \right) \right) \\
&= \frac{Q}{2\pi kL} \ln \left(\frac{r_2}{r_1} \right)
\end{aligned}$$

where we have re-arranged to ensure that $\ln \left(\frac{r_2}{r_1} \right)$ occurs in the expression.

Now transpose to make Q the subject:

$$2\pi kL(\theta_1 - \theta_2) = Q \ln \left(\frac{r_2}{r_1} \right) \quad \implies \quad Q = \frac{2\pi kL(\theta_1 - \theta_2)}{\ln \left(\frac{r_2}{r_1} \right)}$$

as required.

28. The voltage v and current i of an inductor is given by:

$$i = 5(e^{-200t} - e^{-800t}) \quad \text{and} \quad v = e^{-200t} + 400e^{-800t}$$

Find an expression for the power $p = vi$ of the inductor.

Solution:

Expanding the brackets and combining the exponents using the laws of indices:

$$\begin{aligned}
p &= vi \\
&= 5(e^{-200t} - e^{-800t})(e^{-200t} + 400e^{-800t}) \\
&= 5(e^{-200t-200t} - e^{-200t-800t} + 400e^{-800t-200t} - 400e^{-800t-800t}) \\
&= 5(e^{-400t} + 399e^{-1000t} - 400e^{-1600t}) \\
&= 5e^{-1600t}(e^{1200t} + 399e^{600t} - 400)
\end{aligned}$$